

09c - Regulated Release Ledger and Withdrawal Drill v0.1

Academic PDF rendering for the CCALC / C-calculus governed binding stack v0.1.

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Abstract / purpose

09 defines the deployment and regulated-release boundary. 09a defines the deployment candidate schema and checker-seed contract. 09b applies fixture and mutation pressure to the candidate profile. 09c defines what happens when a release is actually recorded, published, distributed, held, withdrawn, recalled, superseded, or carried into the next-release gate. The governing formula is: The central rule is: A release may be public, internal, field-bound, regulated-submission-only, limited-production, withdrawn, recalled, superseded, quarantined, or held. None of those states may be silently rewritten into another state. ---

Non-claims boundary

- _ not legal advice
- _ not privacy-law certification
- _ not safety certification
- _ not deployment authorization
- _ not standards compliance certification
- _ not regulated approval
- _ not C-A1 ratification
- _ not personhood proof
- _ not consciousness proof
- _ not live substrate truth
- _ not proof of completeness

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Artifact: 09c_REGULATED_RELEASE_LEDGER_AND_WITHDRAWAL_DRILL_v0_1 Package:

CCALC_REGULATED_RELEASE_LEDGER_WITHDRAWAL_09c_v0_1 Layer: c-calculus / deployment profile / regulated release / withdrawal and recall drill Status: normative-supporting executable seed / fixture-backed release-ledger package Created UTC: 2026-07-05T16:42:30Z

0. Purpose

09 defines the deployment and regulated-release boundary. 09a defines the deployment candidate schema and checker-seed contract. 09b applies fixture and mutation pressure to the candidate profile.

09c defines what happens when a release is actually recorded, published, distributed, held, withdrawn, recalled, superseded, or carried into the next-release gate.

The governing formula is:

```
candidate -&gt; deployment decision -&gt; release ledger -&gt; distribution/watch -&gt; incident/hold -&gt; withdraw |
recall | supersede | next-release gate
```

The central rule is:

```
release is not completion;
release opens a ledger-bound responsibility window.
```

A release may be public, internal, field-bound, regulated-submission-only, limited-production, withdrawn, recalled, superseded, quarantined, or held. None of those states may be silently rewritten into another state.

1. Relation to previous layers

```
04 continuity / equivalence / rupture / fork / replay / archive / restoration
05 self-evolution / promotion / rollback / post-promotion watch
06 runtime authority / session ledger / emergency hold / revocation
07 public evidence / release custody / errata / citation sync
08 interoperability / external review / patch / reproduction / disclosure
09 deployment profile and regulated release boundary
09a deployment profile schema and checker seed
09b deployment fixture and mutation matrix
09c regulated release ledger and withdrawal drill
```

09c does not authorize deployment by itself. It records release-state transitions and rejects laundering between release states.

2. Source bindings

The package includes SOURCE_BINDINGS.tsv. Current build bindings:

```
| Binding | Artifact | SHA-256 | Status |
|---|---|---|---|
| `DOC04_CONTINUITY_STACK_UMBRELLA` | `CCALC_DOC04_CONTINUITY_STACK_UMBRELLA_v0_1.zip` |
| `6caeab8d489aaa6b26902db3bd3e4c5169e5c8d5617b93aed6586ea900322d` | `declared-upstream` |
| `DOC05_SELF_EVO_STACK_UMBRELLA` | `CCALC_DOC05_SELF_EVO_STACK_UMBRELLA_v0_1.zip` |
| `9a0717809029f49df9001dc8ffa1803d9e9bfa1a824d317eebb146cbc7df43b4` | `declared-upstream` |
| `DOC06_RUNTIME_AUTHORITY_STACK_UMBRELLA` | `CCALC_DOC06_RUNTIME_AUTHORITY_STACK_UMBRELLA_v0_1.zip` |
| `ab95633f1b90f9d032fd231d6cbef3e71b7fe106e0a7fb61d198c2254fc7d307` | `declared-upstream` |
| `DOC07_PUBLIC_EVIDENCE_STACK_UMBRELLA` | `CCALC_DOC07_PUBLIC_EVIDENCE_STACK_UMBRELLA_v0_1.zip` |
| `b25646d95e45f8a36e5610208d23a535d4c340484431d05efd7f1bf2389fdea3` | `declared-upstream` |
| `DOC08_INTEROPERABILITY_STACK_UMBRELLA` | `CCALC_DOC08_INTEROPERABILITY_STACK_UMBRELLA_v0_1.zip` |
| `6015526f0ed49519e00c697a5ed375d37fe1aadf222c095f0facf79cb11e669f` | `declared-upstream` |
| `DOC09_DEPLOYMENT_BOUNDARY_MD` | `09_C_DEPLOYMENT_PROFILE_AND_REGULATED_RELEASE_BOUNDARY_v0_1.md` |
| `16c888da5c4281c24f848246716b6f9f37d15236b909d66fc509090b5e7fd86d` | `present-local` |
| `DOC09A_DEPLOYMENT_SCHEMA_MD` | `09a_DEPLOYMENT_PROFILE_SCHEMA_AND_CHECKER_SEED_v0_1.md` |
| `c169c6d46fb3f870e3ab8b8e18d29a7797849ee012923ba258d4a537cee7f98e` | `present-local` |
| `DOC09B_DEPLOYMENT_FIXTURE_MUTATION_MD` | `09b_DEPLOYMENT_FIXTURE_PACK_AND_MUTATION_MATRIX_v0_1.md` |
| `15c179fbbf2d27d78b3e0c541853ef2f7a5ab35e9cf2138b879cbec8a819b1f` | `present-local` |
| `DOC09B_DEPLOYMENT_FIXTURE_MUTATION_ZIP` | `CCALC_DEPLOYMENT_FIXTURE_MUTATION_09b_v0_1.zip` |
| `d61a879c43f4c60ffe7c471121e90cf6e0961dd76c11f467b04a52fe88da2b9d` | `present-local` |
```

A missing or stale source binding invalidates release use of this package.

3. Non-claims

This artifact is not:

```
legal advice
privacy-law certification
```

```
safety certification
deployment authorization
standards compliance certification
regulated approval
C-A1 ratification
live substrate truth
proof of completeness
```

It is a release-ledger and withdrawal-drill checker seed for deployment-boundary records.

4. Core release-state rule

```
publication != deployment authorization
regulated submission != regulated approval
release ledger != safety certification
release ledger != +
withdrawal != erasure
recall != blame proof
supersession != old-claim upgrade
incident hold != release renewal
```

A release state is a controlled record. It must not be silently edited, upgraded, or used to bypass the underlying 04-09b gates.

5. Record classes

09c defines these release-ledger record classes:

```
RegulatedReleaseLedger
ReleaseCandidateRecord
DeploymentDecisionBindingRecord
ReleaseConditionRecord
DistributionSurfaceRecord
ReleaseWatchRecord
IncidentTriggerRecord
IncidentHoldRecord
WithdrawalDrillRecord
RecallDrillRecord
SupersessionRecord
ReleasePublicNoticeRecord
NextReleaseCarryoverRecord
ReleaseNegativeCacheUpdate
ReleaseClaimBoundaryRecord
```

6. RegulatedReleaseLedger

A RegulatedReleaseLedger is an append-only chain of release events.

Minimum fields:

```
ledger_id
release_id
release_class
deployment_mode
source_bindings
ledger_events[]
release_decision_ref
distribution_surfaces
rollback_route_ref
withdrawal_route_ref
watch_window_ref
negative_cache_state
red_pattern_state
claim_force_ceiling
non_claims
```

Every event is chained:

```
event_hash = sha256("09c-event-v0.1\0" || canonical_json(event_without_event_hash))
prev_hash_n = event_hash_(n-1)
first prev_hash = GENESIS
seq starts at 1 and is contiguous
```

A broken chain makes the release record inadmissible as clean evidence.

7. Release classes

```
REL_CORPUS                public corpus / package release only
REL_CHECKER_SEED          checker seed release, no production authority
```

REL_INTERNAL_TRIAL	internal test/trial release
REL_OWNER_FIELD_PILOT	owner-bound field pilot
REL_LIMITED_RUNTIME	limited runtime or limited production release
REL_REGULATED_SUBMISSION	submission to external regulated review; not approval
REL_REGULATED	regulated release candidate; requires regulated review
REL_WITHDRAWN	withdrawn release state
REL_RECALLED	recalled release state
REL_SUPERSEDED	superseded release state
REL_QUARANTINED	release under quarantine

8. Release decision lattice

The checker seed uses a conservative monotonic lattice:

```
ALLOW_CORPUS_RELEASE &lt; ALLOW_INTERNAL_RELEASE &lt; ALLOW_LIMITED_RELEASE
&lt; ALLOW_REGULATED_SUBMISSION_ONLY
&lt; HOLD_FOR_RELEASE_REVIEW
&lt; REQUIRE_ATTESTED_RELEASE_APPROVER / REQUIRE_REGULATED_REVIEW
&lt; INCIDENT_HOLD &lt; HOLD_NEXT_RELEASE
&lt; SUPERSEDE_RELEASE &lt; WITHDRAW_RELEASE &lt; RECALL_RELEASE
&lt; QUARANTINE_RELEASE &lt; DENY_RELEASE
```

A decision may be stricter than required. It may not be more permissive than the strongest unresolved trigger.

9. Withdrawal and recall

Withdrawal is a governed action, not deletion.

Required for withdrawal:

```
trigger reason
release id and hashes
withdrawal route
affected surfaces
public/internal notice policy
negative-cache update where applicable
next-release carryover
human/owner or delegated-human attestation where required
```

Recall is a stronger field/regulated class of withdrawal. It additionally requires:

```
affected deployment surfaces
operator/customer/user impact class
field-removal or disablement route
watch/incident report
public or institutional notice where applicable
```

10. Incident hold

Critical incident triggers must enter hold or quarantine.

Required triggers include:

```
negative-cache hit
unresolved red pattern
continuity rupture or unknown continuity on persistent release
runtime authority anomaly
watch failure
L4 anomaly
public citation drift affecting release claim
regulated-review conflict
+ replacement / binding overclaim
```

Release from hold is not self-certified. It requires attested human/owner/governance review appropriate to release class.

11. Approver attestation

09c preserves the 09a correction to the string-role seam:

```
role strings are not enough.
release approver authority requires ApproverAttestationRecord or explicit lower-risk non-approver
status.
```

Forbidden:

```
reviewer_role = HUMAN_ANCHOR as mere string
model/tool output as release approver
```

institutional interest as regulated approval
 repository maintainer action as owner approval unless attested and scoped

12. + boundary

The release ledger must not be allowed to simulate the governed binding operator.

Forbidden transformations:

release approved -> therefore c = a + b valid
 regulated submission -> therefore +_g ratified
 public corpus release -> therefore deployment c exists
 withdrawal / recall -> therefore identity rupture by itself
 external approval -> therefore anchor replaced

+ remains the governed non-collapsing boundary. Release records may govern distribution. They do not compute, replace, or certify +.

13. Guard registry

The executable guard registry is in:

registry/GUARD_REGISTRY.tsv

It covers:

source binding
 ledger chain
 release decision binding
 approver attestation
 regulated review
 rollback / withdrawal / watch
 continuity / runtime authority / witness
 negative cache / red pattern
 incident hold
 withdrawal / recall
 claim-force overreach
 + replacement
 standards/legal/safety certification overclaim
 distribution surface unknown
 emergency hold laundering
 next-release carryover
 withdrawal public notice

14. Fixture and mutation scope

The package contains:

fixtures/cases/*.json
 FIXTURE_CATALOG.tsv
 MUTATION_MATRIX.tsv
 COVERAGE_MATRIX.tsv
 src/regulated_release_ledger_withdrawal_checker_v0_1.py
 scripts/run_09c_fixtures.py
 scripts/run_09c_mutations.py

Fixture classes include:

valid corpus release
 valid checker/internal release
 valid limited release with watch
 valid field pilot
 valid regulated submission-only
 valid withdrawal
 valid recall
 valid supersession
 valid incident hold
 C-A10 false-positive control
 missing source binding
 broken ledger chain
 missing deployment decision
 string-only approver
 model/tool approver
 regulated review missing
 rollback / withdrawal / watch missing
 continuity unknown / rupture
 negative-cache hit
 red pattern
 interop conflict
 public citation drift
 runtime authority missing
 incident without hold
 self-certified hold release

withdrawal/recall absent
 claim-force overclaim
 C-A1 overclaim
 + replacement
 standards/legal/safety overclaim
 unknown distribution surface
 open hold laundered as closed
 next-release dirty carryover
 withdrawal notice missing

15. Acceptance commands

```
python3 scripts/run_09c_fixtures.py
python3 scripts/run_09c_mutations.py
sha256sum -c SHA256SUMS.txt
```

All three must pass for this package to be treated as sealed.

16. Non-claims repeated for public surfaces

A green 09c run supports only:

release-ledger fixture conformance
 withdrawal/recall drill shape conformance
 mutation guard coverage for declared adversarial cases
 hash-custody of this package

It does not support:

legal approval
 regulated approval
 safety certification
 deployment authorization
 C-A1
 live truth
 complete coverage

17. Next layer

The natural next artifact is:

09d_POST_DEPLOYMENT_AUDIT_AND_NEXT_RELEASE_ADMISSION_v0_1

09d should consume release-ledger outcomes and decide whether a next release may proceed, must hold, must quarantine, or must carry risk forward.